C Programming – Pointers

1. Declare an integer, store its address in a pointer, and print the variable's value three ways: directly, via dereferencing the pointer, and via &variable.
2. Take two integers as input, and using only pointers swap their values. *(Write a function swap(int *a, int *b)).*
3. Given an array of 10 integers, use a pointer (not indexing) to print every element.
4. Use a pointer to find the **sum** and **average** of all elements in an integer array.
5. Use a pointer to find the **maximum** and **minimum** value in an array, without using arr[i] anywhere.
6. Given a pointer p pointing to arr[0] of an int array, predict the address p + 3 will hold, then verify with printf("%p", ...).
7. Reverse an array in place using two pointers — one starting at the first element, one at the last, swapping and moving inward until they meet.
8. Write a program using pointer arithmetic to count how many elements in an array are divisible by 5.